

Containers

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Accessibility Notice

This document is also available in HTML format at:

Link: <https://aholdengouveia.name/LinuxAdmin/labexercices/containers.html>

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Objectives:

- The objective of this lab is to introduce students to containerization technologies, including Docker and Kubernetes, which are widely used in modern server administration and application deployment. Through hands-on installation and configuration exercises, students will gain practical experience deploying containerized applications, understand the advantages and limitations of containers, and evaluate their use in enterprise environments.

Complete the following problems

References, a video, a PowerPoint and some notes are available at my website
Link: <https://www.aholdengouveia.name/LinuxAdmin/containers.html>

You may do this on one of your servers.

1. Install Docker and Kubernetes on your server.
You can use Microk8s as your Kubernetes installation method
Instructions: Link: <https://microk8s.io/>
2. Go through the MiniKube tutorial
Tutorial: Link: <https://kubernetes.io/docs/tutorials/hello-minikube/>
3. Get a basic web server working using Docker OR kubernetes on ONE of your servers.
You can grab the docker image from Link: https://hub.docker.com/_/httpd
I also have some instructions and sample yaml files available for you on my website Link: <https://www.aholdengouveia.name/SmartHome/Virtualizationsetup.html>
If you would prefer you can substitute another server for Apache such as Home Assistant or ZoneMinder.

Deliverables

Target Market is someone technically inclined but unfamiliar with virtualization. Include screenshots to walk them through the install process, as well as explanations of what you're doing and any troubleshooting that was needed.

1. A document on how to install Docker and Kubernetes on your server.
2. A short document describing how the Minikube tutorial went, what gave you trouble (if anything) and how you fixed it. Include screenshots to show how it went and what steps you did. Include screenshots of Minikube up and running.
3. An executive summary on what containers are, pro/cons of containers, and why you think Acme corp should or shouldn't consider using them going forward. Executive summary should be less than 1 page.
4. A link to resources for containers that you have found helpful that I haven't already shared anywhere.